Electronic Journal of Statistics
ISSN: 1935-7524

The Adaptive Elastic Horseshoe Prior: Data-Driven Regularisation for Mixed Sparse-Dense Feature Structures via Momentum-Based Empirical Bayes

George Paul[1] **and Michael Tipping**[1]

[1] *Department of Computer Science, University of Bath, Bath BA2 7AY, UK, e-mail:* gpaul2002@gmail.com

Abstract: We introduce the *Adaptive Elastic Horseshoe* (AEH) prior, a global-local shrinkage prior for Bayesian linear regression that learns the balance between sparsity-inducing and grouping regularisation from data. Existing shrinkage priors impose a fixed regularisation geometry: the horseshoe does not accommodate correlated dense signals, while the elastic net fixes the L1/L2 mixing ratio. The AEH prior addresses this by augmenting horseshoe local scales with an adaptive elastic net component whose mixing parameter α and strength β are updated via momentum-driven empirical Bayes steps. We show the updates correspond to approximate EM on a mean-field variational lower bound and prove the AEH prior achieves minimax-optimal posterior contraction for the sparse normal means problem. Fitted via Hamiltonian Monte Carlo within a group-wise ARD framework, the AEH prior is evaluated on eight simulation experiments and on the Efron et al. (2004) diabetes dataset ($n = 442$, $p = 10$). Decision boundary analyses at $p = 20$ and $p = 200$ show AEH outperforms the horseshoe in 25 of 30 configurations, with gains where $\rho \gtrsim 0.6$ and group size ≥ 5. The HMC sampler achieves substantially higher Effective Sample Size under AEH than under the standalone horseshoe (mean ESS 692.8 vs. 395.0), reflecting a better-conditioned posterior geometry. Limitations including convergence failures at low n/p and over-wide predictive intervals in dense settings are reported transparently.

MSC2020 subject classifications: Primary 62F15, 62J07; secondary 65C05, 62C10.
Keywords and phrases: Bayesian shrinkage priors, horseshoe prior, elastic net, adaptive regularisation, empirical Bayes, Hamiltonian Monte Carlo, variational inference, posterior contraction, sparse normal means..

1. Introduction

Shrinkage priors for high-dimensional Bayesian regression have received sustained attention over the past two decades [3, 14, 2, 15]. A common thread in this literature is the *global-local* representation [15]: a coefficient w_j is assigned a zero-mean Gaussian with variance $\lambda_j^2\tau^2$, where τ is a global shrinkage parameter shared across all coefficients and λ_j is a feature-specific local scale. Different choices of prior on λ_j recover familiar estimators: half-Cauchy local scales yield the horseshoe [3]; exponential local scales yield the Bayesian lasso [8]; point-mass mixtures yield spike-and-slab [11].

Despite this theoretical richness, two structural limitations affect the entire class. First, the local scales λ_j are inferred independently for each feature: the prior encodes no information about whether features are likely to form correlated groups requiring ridge-like regularisation or sparse singletons requiring lasso-like regularisation. Second, the *global* shrinkage parameter τ is either fixed by the user or inferred under a hyperprior, but in either case it applies uniform global pressure that cannot simultaneously accommodate dense correlated blocks and sparse irrelevant features within the same model.

These limitations are consequential whenever the regression design matrix $\boldsymbol{X}$ exhibits what we term *mixed sparse-dense* structure: a subset of features that are strongly correlated with each other and collectively predictive (requiring L2-like grouping to avoid coefficient instability), alongside a larger set of weakly relevant or irrelevant features (requiring aggressive L1-like pruning). Such structure arises naturally in many scientific and applied settings, including genomics (dense gene sets alongside sparse causal variants), finance (correlated factor exposures alongside sparse idiosyncratic signals), and environmental modelling (dense sub-metered consumption signals alongside sparse structural predictors).

The elastic net [18] addresses the correlated-feature problem in a frequentist setting by mixing L1 and L2 penalties, but the mixing ratio $\alpha \in [0, 1]$ is a fixed hyperparameter. Cross-validated selection of α is computationally expensive and ignores uncertainty in the regularisation geometry. The Bayesian elastic net [8, 9] places priors on the penalty parameters, but these priors are static and cannot adapt to the observed importance and correlation structure of the data.

Contribution. We introduce the *Adaptive Elastic Horseshoe* (AEH) prior, which addresses the gap above through three components:

1. **Adaptive elastic augmentation.** The horseshoe local scales λ_j are augmented with a momentum-updated elastic net penalty whose L1/L2 mixing ratio α and strength β are updated at each iteration of HMC inference based on observed feature importance and prediction uncertainty ratios.
2. **Variational EM grounding.** We show that the update rules for α and β are approximate M-steps in an EM algorithm maximising a mean-field variational lower bound on the log marginal likelihood. This establishes the adaptive updates as a principled Type-II maximum likelihood procedure rather than a heuristic.
3. **Group-wise ARD embedding.** The AEH prior is assigned to feature groups expected to exhibit dense correlated structure; groups expected to be sparse receive a hierarchical ARD prior. This group-wise architecture enables differential regularisation aligned with prior structural knowledge.

We establish theoretical properties including minimax-optimal posterior contraction (Theorem 4.5) and fixed-point convergence of the adaptive updates (Proposition 4.1), and validate the prior on eight controlled simulations and a real dataset.

Scope of theory. Theorem 4.5 is established under the sparse normal means model with an identity design matrix. This is the standard setting for analysing global-local shrinkage priors [16, 2], but it does not directly cover the correlated design setting that motivates the AEH prior. Extending posterior contraction theory to correlated designs is a non-trivial open problem; we address the correlated setting empirically in Section 6 and discuss the theoretical gap in Section 8.4.

Organisation. Section 2 reviews global-local shrinkage priors and the elastic net. Section 3 defines the AEH prior formally. Section 4 establishes the variational EM interpretation and contraction results. Section 5 describes the HMC inference procedure. Section 6 presents synthetic experiments. Section 7 describes the diabetes application. Section 8 discusses limitations and extensions. Section 9 concludes.

2. Background and Related Work

2.1. Global-Local Shrinkage Priors

The global-local framework [15] represents a regression coefficient w_j as

$$w_j \mid \lambda_j, \tau \sim \mathcal{N}(0, \lambda_j^2 \tau^2), \qquad \lambda_j \sim \pi(\lambda_j), \qquad \tau \sim \pi(\tau). \tag{1}$$

The marginal prior on w_j obtained by integrating out λ_j is heavy-tailed when $\pi(\lambda_j)$ has mass near zero and near infinity, producing the characteristic "nearly black" behaviour: small coefficients are shrunk aggressively toward zero while large coefficients escape shrinkage almost entirely.

The **horseshoe prior** [3] sets $\lambda_j \sim C^+(0, 1)$ (half-Cauchy). The regularised horseshoe [14], which places a Student-t slab on the non-zero component to avoid excessive shrinkage of large signals; this is arguably the current state of the art among global-local priors and forms our primary baseline.

The **Dirichlet-Laplace** prior [2] achieves minimax-optimal contraction by coupling local scales through a Dirichlet distribution, inducing borrowing across features. The **spike-and-slab** prior [11] provides exact variable selection via a point-mass mixture but is computationally prohibitive at high dimensions due to the 2^p model space.

A key limitation shared by all of the above is that the regularisation *geometry* - the balance between sparsity-inducing and grouping behaviour - is fixed by the prior specification. When features are correlated, the horseshoe can exhibit instability analogous to the lasso's arbitrary selection among correlated predictors [18]. No existing prior in the global-local family adapts its geometry based on the observed correlation and importance structure of the data.

2.2. The Elastic Net and Its Bayesian Analogues

The elastic net [18] regularises via the combined penalty

$$\Omega(\boldsymbol{w}) = \alpha \|\boldsymbol{w}\|_1 + (1 - \alpha)\|\boldsymbol{w}\|_2^2, \tag{2}$$

with $\alpha \in [0, 1]$ mixing lasso ($\alpha = 1$) and ridge ($\alpha = 0$) behaviour. For correlated predictors, $\alpha < 1$ tends to select groups of variables together (the "grouping effect") while still inducing sparsity. However, α must be specified or cross-validated; it does not adapt to the data during fitting.

The Bayesian lasso [8] places a double-exponential prior on coefficients, equivalent to a Laplace (L1) penalty in the MAP estimate. Li and Lin [9] extend this to a Bayesian elastic net by combining Laplace and Gaussian priors with fixed mixing weights. Neither formulation allows the mixing ratio to vary adaptively.

2.3. Empirical Bayes and Adaptive Hyperparameters

Type-II maximum likelihood [10], or empirical Bayes, maximises the log marginal likelihood $\log p(\boldsymbol{y} \mid \boldsymbol{\theta})$ with respect to hyperparameters $\boldsymbol{\theta}$. In the ARD framework [10, 12], feature-specific precision parameters are updated iteratively via EM, pruning irrelevant features by driving their precisions to infinity. Our adaptive updates for α and β can be understood as an extension of this idea to the elastic mixing geometry: the EM M-step not only updates shrinkage strength but also the functional form of the penalty, steering between L1 and L2 behaviour based on the model's own posterior summaries. We formalise this connection in Section 4.

2.4. *Positioning of the AEH Prior*

Table 1 summarises how the AEH prior relates to existing methods along the key dimensions of sparsity, grouping, adaptivity, and calibrated uncertainty.

Table 1

Comparison of shrinkage priors along key dimensions. ✓ = supported natively; ○ = partially or approximately; × = not supported.

Prior	Sparsity	Grouping	Adaptive geometry	Calibrated UQ
Horseshoe [3]	✓	×	×	✓
Reg. Horseshoe [14]	✓	○	×	✓
Dirichlet-Laplace [2]	✓	○	×	✓
Bayesian Lasso [8]	✓	×	×	✓
Bayesian Elastic Net [9]	○	✓	×	✓
Spike-and-Slab [11]	✓	×	×	✓
AEH (proposed)	✓	✓	✓	✓

3. The Adaptive Elastic Horseshoe Prior

3.1. *Model Specification*

Consider the standard Bayesian linear regression model

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{w} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I}_n), \tag{3}$$

where $\boldsymbol{y} \in \mathbb{R}^n$, $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, $\boldsymbol{w} \in \mathbb{R}^p$, and $\sigma^2 > 0$. Let $\mathcal{G} = \{G_1, \ldots, G_K\}$ be a partition of the feature index set $\{1, \ldots, p\}$ into K groups based on prior structural knowledge. Group G_k receives prior $\pi_k(\cdot \mid \boldsymbol{\theta}_k)$.

In our group-wise architecture, group G_1 (features expected to be dense and correlated) receives the AEH prior; remaining groups receive the Hierarchical ARD prior (Section 3.5).

3.2. *The AEH Prior*

Definition 3.1 (AEH Prior)**.** Let $j \in G_1$. The AEH prior on w_j is the global-local Gaussian

$$w_j \mid \lambda_j, \tau \;\sim\; \mathcal{N}\left(0,\; \frac{\lambda_j^2 \tau^2}{1 + \beta \cdot e_j(\alpha)}\right), \tag{4}$$

where $\tau > 0$ is a global shrinkage parameter, $\lambda_j > 0$ is a feature-specific local scale, $\alpha \in [0, 1]$ is an L1/L2 mixing parameter, $\beta > 0$ controls the overall elastic penalty influence, and

$$e_j(\alpha) \;=\; \alpha |w_j| + (1 - \alpha) w_j^2 \tag{5}$$

is the elastic net penalty term for feature j. The global and local scales receive the standard horseshoe hyperpriors:

$$\tau \sim C^+(0, \tau_0), \qquad \lambda_j \sim C^+(0, 1), \tag{6}$$

where $\tau_0 = p_0/(p - p_0) \cdot \sigma/\sqrt{n}$ with p_0 the prior expected number of non-zero coefficients [14].

Remark 3.2. When $\beta = 0$ the AEH prior reduces to the standard horseshoe. When $\beta > 0$ and $\alpha = 0$ it reduces to a horseshoe with an additional L2 penalty on w_j. When $\alpha = 1$ it adds an L1 penalty, concentrating mass more aggressively at zero. The AEH prior therefore nests the standard horseshoe and provides a continuum of regularisation geometries parameterised by (α, β).

3.3. Local Scale Update

The local scale λ_j is updated via a momentum-based gradient step. Define the effective horseshoe scale as

$$s_j = \frac{1}{w_j^2/(2\tau^2) + \beta \cdot e_j(\alpha) + \epsilon}, \qquad \epsilon = 10^{-6}, \tag{7}$$

and the gradient

$$g_j = -s_j + \beta \cdot e_j(\alpha). \tag{8}$$

The momentum update is

$$m_j^{(t+1)} = \rho\, m_j^{(t)} + g_j^{(t)}, \tag{9}$$

$$\lambda_j^{(t+1)} = \mathrm{clip}\Big(\lambda_j^{(t)} + m_j^{(t+1)},\ \epsilon,\ \infty\Big), \tag{10}$$

where $\rho \in [0, 1)$ is a momentum decay parameter (default $\rho = 0.9$).

3.4. Adaptive Hyperparameter Updates

The key innovation is the data-driven update of α and β. Let $\hat{w}_j = \mathbb{E}[w_j \mid \boldsymbol{y}]$ denote the posterior mean of coefficient j (approximated by the HMC sample mean). Define the *importance ratio*

$$r_\alpha^{(t)} = \frac{\sum_{j=1}^{p} \mathbf{1}[|\hat{w}_j| > \delta]}{p}, \tag{11}$$

the fraction of features whose posterior mean exceeds a threshold $\delta > 0$ (default $\delta = 0.01$; sensitivity analysis in the application shows results are robust to this choice), and the *uncertainty ratio*

$$r_\beta^{(t)} = \frac{\mathrm{RMSE}_{\mathrm{train}}^{(t)}}{\mathrm{RMSE}_{\mathrm{train}}^{(0)}}, \tag{12}$$

the ratio of current to initial training RMSE. The hyperparameter updates are

$$\alpha^{(t+1)} = \mathrm{clip}\Big(\alpha^{(t)} + \gamma\Big(0.5 - r_\alpha^{(t)}\Big),\ 0.1,\ 0.9\Big), \tag{13}$$

$$\beta^{(t+1)} = \mathrm{clip}\Big(\beta^{(t)} + \gamma\Big(1.0 - r_\beta^{(t)}\Big),\ 0.1,\ 10.0\Big). \tag{14}$$

The learning rate $\gamma > 0$ has default value $\gamma = 0.01$. The bounds $[0.1, 0.9]$ for α deliberately exclude the degenerate endpoints $\{0, 1\}$: pure ridge ($\alpha = 0$) eliminates sparsity entirely and pure lasso ($\alpha = 1$) reintroduces instability under correlated features; clipping to a strict interior interval avoids numerical singularities in the penalty gradient $\partial e_j / \partial \alpha$ and ensures the elastic net term remains well-defined throughout inference.

Intuitively, α is steered toward 1 (L1-dominant, sparse) when few features are active ($r_\alpha < 0.5$) and toward 0 (L2-dominant, grouping) when many features are active. Similarly, β increases when the model has high prediction error and decreases when error is low. We provide formal grounding for these updates in Section 4.

3.5. Hierarchical ARD Prior

For groups $G_2, \ldots, G_K$ (sparse feature groups), each coefficient w_j receives a zero-mean Gaussian with feature-specific precision α_j inferred from data:

$$w_j \mid \alpha_j \ \sim\ \mathcal{N}(0, \alpha_j^{-1}), \qquad \alpha_j \ \sim\ \mathrm{Gamma}(a_0, b_0), \tag{15}$$

with $a_0 = b_0 = 10^{-4}$ (weakly informative). The ARD update in the EM M-step is

$$\alpha_j^{\text{new}} = \frac{1}{\mathbb{E}[w_j^2]}, \tag{16}$$

which drives $\alpha_j \to \infty$ (effective removal) when $\mathbb{E}[w_j^2] \approx 0$.

4. Theoretical Properties

4.1. Variational EM Interpretation of the Adaptive Updates

We establish that the adaptive updates in Equations (13) and (14) are approximate M-steps in an EM algorithm maximising a mean-field variational lower bound on the log marginal likelihood $\log p(\boldsymbol{y})$.

Variational lower bound. Let $q(\boldsymbol{w}) = \prod_j q_j(w_j)$ be a mean-field variational approximation to the posterior $p(\boldsymbol{w} \mid \boldsymbol{y})$. The Evidence Lower BOund (ELBO) is

$$\mathcal{L}(q, \boldsymbol{\theta}) \ =\ \mathbb{E}_q[\log p(\boldsymbol{y} \mid \boldsymbol{w})] - \mathrm{KL}(q(\boldsymbol{w}) \,\|\, p(\boldsymbol{w} \mid \boldsymbol{\theta})), \tag{17}$$

where $\boldsymbol{\theta} = (\alpha, \beta, \tau, \{\lambda_j\})$ collects the AEH hyperparameters. Since $\mathcal{L}(q, \boldsymbol{\theta}) \le \log p(\boldsymbol{y})$ with equality when $q = p(\cdot \mid \boldsymbol{y})$, maximising the ELBO over $\boldsymbol{\theta}$ at fixed q yields an ascent on the marginal likelihood.

M-step for α. Treating $q_j(w_j) = \mathcal{N}(w_j; \mu_j, \nu_j)$ (Gaussian variational family), the KL term involving α is

$$-\mathrm{KL}_\alpha = -\tfrac{1}{2}\sum_j \log(1+\beta\cdot e_j(\alpha)) - \tfrac{1}{2}\sum_j \frac{\mu_j^2+\nu_j}{\lambda_j^2\tau^2/(1+\beta\cdot e_j(\alpha))}. \tag{18}$$

The gradient with respect to α is

$$\frac{\partial\mathcal{L}}{\partial\alpha} = -\tfrac{1}{2}\sum_j \frac{\beta\cdot\frac{\partial e_j}{\partial\alpha}}{1+\beta e_j}\left(1-\frac{\mu_j^2+\nu_j}{\lambda_j^2\tau^2/(1+\beta e_j)}\right), \tag{19}$$

where $\partial e_j/\partial\alpha = |w_j| - w_j^2$. A stochastic gradient ascent step with step size γ using HMC-approximated posterior means $\mu_j \approx \hat{w}_j$ yields an update of the form in Equation (13).

The quantity $(0.5 - r_\alpha^{(t)})$ is a computable proxy for the expectation of Equation (19) over the feature set. To see this, note that Equation (19) is negative (pushing α down toward L2) when features have large posterior means relative to their prior variance, which occurs precisely when many features are active ($r_\alpha > 0.5$). Conversely, the gradient is positive (pushing toward L1) when few features are active. The proxy $(0.5 - r_\alpha)$ thus has the correct sign and qualitative magnitude, making it a valid stochastic surrogate for the exact gradient.

M-step for β. An analogous argument applies to β. The gradient $\partial\mathcal{L}/\partial\beta$ is positive (pushing β up, strengthening regularisation) when the current model underfits the data, which is reflected in high RMSE. The uncertainty ratio $r_\beta^{(t)}$ approximates this gradient: when $r_\beta > 1$ (RMSE has increased relative to initialisation), $(1 - r_\beta) < 0$ and β decreases; when RMSE is high relative to initialisation, $(1 - r_\beta)$ remains negative, increasing β only when current RMSE is below its initial value, indicating the model has sufficient fit and can reduce regularisation.

Proposition 4.1 (Approximate M-step convergence). *Let $\{\alpha^{(t)}, \beta^{(t)}\}_{t\geq 0}$ be the sequence generated by Equations (13) and (14) with step size $\gamma > 0$ and clipping to $[\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}]$. Suppose $\mathcal{L}(q, \alpha, \beta)$ is continuously differentiable and bounded above in (α, β). Then the sequence $\{\mathcal{L}(q, \alpha^{(t)}, \beta^{(t)})\}$ converges (in expectation over the stochastic gradient noise) to a stationary point of the ELBO on the constraint set.*

Proof. See Section A for the full proof. □

Remark 4.2. Theorem 4.1 guarantees convergence to a *local* stationary point. Global convergence is not claimed; the ELBO may have multiple local maxima corresponding to different regularisation geometries. In practice, sensitivity analysis in the application shows results are robust to initialisation of α and β.

4.2. Posterior Contraction

We analyse the AEH prior under the sparse normal means model: $y_j = w_j + \varepsilon_j$, $\varepsilon_j \overset{iid}{\sim} \mathcal{N}(0,1)$, $j = 1, \ldots, p$, with $\boldsymbol{w}_0$ having at most s_0 non-zero entries. We require two lemmas before stating the main result.

Lemma 4.3 (Effective local scale)**.** *Under the AEH prior, the effective local scale satisfies* $\tilde{\lambda}_j := \sqrt{\kappa_j}\lambda_j \le \lambda_j$*, where* $\kappa_j = (1 + \beta e_j(\alpha))^{-1} \in (0, 1]$*. For zero components (*$w_{0j} = 0$*) and* $\beta > 0$*, we have* $\kappa_j \le (1 + \beta\alpha|w_j|)^{-1} < 1$*, so the effective local scale is strictly smaller than under the standard horseshoe.*

Proof. Since $e_j(\alpha) = \alpha|w_j| + (1 - \alpha)w_j^2 \ge 0$ for all $\alpha \in [0, 1]$ and $\beta > 0$, we have $\kappa_j = (1 + \beta e_j(\alpha))^{-1} \le 1$, and hence $\tilde{\lambda}_j = \sqrt{\kappa_j}\lambda_j \le \lambda_j$. For $w_{0j} = 0$, near the origin we have $e_j(\alpha) > 0$ for any $\alpha > 0$, giving $\kappa_j < 1$ strictly. □

Lemma 4.4 (Prior mass on ℓ_2 balls)**.** *For the AEH prior with global scale* $\tau = \tau_0 = s_0/\sqrt{p \log p}$*, there exist constants* $C_1, C_2 > 0$ *(depending on* β, α*) such that for any* $\boldsymbol{w}_0 \in \ell_0[s_0]$*,*

$$\Pr\Big(\|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2 \le C_1 s_0 \log(p/s_0)\Big) \ \ge\ \exp(-C_2 s_0 \log(p/s_0)).$$

Proof. For signal components ($j \in S$, $|w_{0j}| \ge c > 0$), the heavy-tailed half-Cauchy prior on λ_j ensures that $\lambda_j^2 \gg \beta e_j$ with probability at least $1 - O(1/p)$ for $|w_{0j}|$ bounded below. Specifically, since $\lambda_j \sim C^+(0, 1)$, we have $\Pr(\lambda_j > M) = 2/\pi \cdot \arctan(1/M) \sim 2/(\pi M)$ for large M, which is polynomially heavy. Setting $M = (\beta e_j)^{1/2}$ gives $\Pr(\lambda_j^2 > \beta e_j) \ge 1 - O(\beta^{1/2}|w_{0j}|^{1/2}/p)$, which is bounded away from zero for fixed β and $|w_{0j}|$ bounded. Thus for signal components $\kappa_j \to 1$ with high probability, and the AEH prior behaves like the horseshoe. The horseshoe places sufficient mass on ℓ_2 balls at the rate $\exp(-C_2 s_0 \log(p/s_0))$ by Lemma 3 of [16]; the same bound applies to the AEH prior with constants that depend continuously on (α, β). □

Theorem 4.5 (Posterior contraction under AEH prior)**.** *Let* $p(\boldsymbol{w})$ *be the AEH prior with global scale* $\tau = \tau_0 = s_0/\sqrt{p \log p}$ *and* $\beta > 0$*,* $\alpha \in [0, 1]$*. Then the posterior satisfies*

$$\mathbb{E}_{\boldsymbol{w}_0}\Big[\Pi\Big(\|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2 \ge M_n s_0 \log(p/s_0) \mid \boldsymbol{y}\Big)\Big] \to 0 \tag{20}$$

as $n \to \infty$*, for any* $M_n \to \infty$*, uniformly over* $\ell_0[s_0] = \{\boldsymbol{w} : \|\boldsymbol{w}\|_0 \le s_0\}$*.*

Proof. See Section B for the full proof. □

Corollary 4.6. *The AEH prior achieves the same minimax-optimal posterior contraction rate as the standard horseshoe prior, regardless of* $\alpha \in [0, 1]$ *and* $\beta \ge 0$*.*

Remark 4.7. Theorem 4.5 is established under the sparse normal means model with an identity design matrix. Extension to correlated designs is non-trivial: the effective local scales $\tilde{\lambda}_j = \sqrt{\kappa_j}\lambda_j$ depend on w_j, which in turn depends on $\boldsymbol{X}^\top\boldsymbol{X}$ in the correlated setting, making the three conditions of [16] harder to verify. This constitutes an open theoretical problem; the correlated setting is addressed empirically in Section 6.

5. Inference via Hamiltonian Monte Carlo

The AEH model is fitted using Hamiltonian Monte Carlo [5, 13].

HMC configuration. Four parallel chains are run with 2,000 samples each (1,000 warmup), step size $\epsilon = 0.01$, 10 leapfrog steps, and a target acceptance rate of 0.65. The adaptive hyperparameter updates (Section 3.4) are applied at each warmup iteration, with (α, β) held fixed during the post-warmup sampling phase to ensure valid Markov chain transitions.

Convergence diagnostics. Convergence is assessed via Gelman-Rubin $\hat{R}$ statistics [7] (threshold $\hat{R} \leq 1.05$ for Experiments A-C due to their higher computational demands; $\hat{R} \leq 1.01$ for all other experiments) and Effective Sample Size (ESS; threshold ESS > 500 per parameter). We report bulk-ESS and tail-ESS separately following [17]. In Experiments A-C, we apply a *convergence gate*: replications failing the $\hat{R}$ threshold have all performance metrics (MSE, FPR, ESS, PICP) set to `NaN` and are excluded from summary statistics. Convergence rates are reported alongside all results.

Posterior predictive distribution. Prediction intervals are drawn from the posterior predictive distribution

$$p(\tilde{y} \mid \boldsymbol{y}) = \int p(\tilde{y} \mid \boldsymbol{w}, \sigma^2)\, p(\boldsymbol{w}, \sigma^2 \mid \boldsymbol{y})\, d\boldsymbol{w}\, d\sigma^2, \tag{21}$$

approximated via Monte Carlo integration over the HMC samples. Calibration is assessed using Prediction Interval Coverage Probability (PICP) at nominal levels $\{50\%, 80\%, 90\%, 95\%, 99\%\}$.

Algorithm 1 summarises the full inference procedure.

Algorithm 1 AEH-HMC Inference

Require: Data $(\boldsymbol{X}, \boldsymbol{y})$, group partition $\mathcal{G}$, initial hyperparameters $(\alpha^{(0)}, \beta^{(0)}, \tau^{(0)})$, HMC configuration.

- **Initialise** $\lambda_j^{(0)} \leftarrow 1$, $m_j^{(0)} \leftarrow 0$ for all $j \in G_1$.
- **for** $t = 1, \ldots, T_{\text{warmup}}$ **do**
  - Draw HMC proposal for $\boldsymbol{w}^{(t)}$, $\tau^{(t)}$, $\{\lambda_j^{(t)}\}$ using current hyperparameters.
  - Compute $r_\alpha^{(t)}, r_\beta^{(t)}$ from $\boldsymbol{w}^{(t)}$ and training RMSE.
  - Update $\alpha^{(t+1)}, \beta^{(t+1)}$ via Equations (13) and (14).
  - Update $\lambda_j^{(t+1)}$ via Equations (9) and (10).
- **end for**
- **for** $t = T_{\text{warmup}} + 1, \ldots, T$ **do**
  - Draw HMC samples with (α, β) fixed at $(\alpha^{(T_{\text{warmup}})}, \beta^{(T_{\text{warmup}})})$.
- **end for**
- **return** posterior samples $\{\boldsymbol{w}^{(t)}\}_{t > T_{\text{warmup}}}$ and converged hyperparameters (α^*, β^*).

6. Simulation Studies

We conduct eight controlled simulation experiments to validate the AEH prior's adaptive behaviour on known ground truth. Table 2 summarises all experiments. Experiments 1–3 use $n = 300$, $p = 20$ and serve as the primary validation suite; Experiment 4 extends to a high-dimensional regime. Experiments A-C are supplementary stress tests exploring joint scaling, group prior misspecification, and a genomics-like design respectively; these apply a convergence gate ($\hat{R} \leq 1.05$) and results are reported only for converged replications. All experiments use 50 independent replications unless noted.

6.1. Experiment 1: Sparse Normal Means - Sanity Check

Setup. We simulate $w_{0j} = 3$ for $j \in \mathcal{S}$ and $w_{0j} = 0$ otherwise, where $|\mathcal{S}| = s_0 \in \{2, 5, 10\}$ and $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_p)$ (independent features). We compare AEH, standard horseshoe, regularised horseshoe [14], Bayesian elastic net [9], and ARD.

Table 2

Overview of all simulation experiments. Experiments A-C apply an $\hat{R} \leq 1.05$ convergence gate; convergence rates are reported alongside results.

Experiment	Design	Replications	Primary question
Exp 1	Sparse normal means, $s_0 \in \{2, 5, 10\}$	50	Sanity check: sparsity recov
Exp 2	Correlated block, $\rho \in \{0.5, 0.7, 0.9\}$	50	Adaptive α^* under correlatio
Exp 3	ESS vs $n \in \{100, \ldots, 2000\}$	20	Sampling efficiency scaling
Exp 4	High-dim $p = 200$, $n \in \{500, 1000, 2000\}$	50	Scaling at adequate n/p
Exp D	Decision boundary ($p = 20$), $\rho\times$ group_size grid	30/cell	When does AEH beat horse
Exp D′	Decision boundary ($p = 200$, $n = 600$)	10/cell	Extends heuristic to modera
Exp A	Joint $p \times \rho$ stress, $p \in \{50, 200, 500\}$	30/cell	Scaling limits
Exp B	Group prior misspecification	50/condition	Robustness to prior assignm
Exp C	Genomics-like, $n = 500$, $p = 300$	30	Scalability to higher p

Results. Table 3 reports mean signal MSE and FPR across 50 replications. This experiment serves primarily as a sanity check: on a sparse independent design the AEH prior should not improve over the standard horseshoe, and it does not. Fixed-shrinkage methods (horseshoe, regularised horseshoe, ARD) achieve lower signal MSE than AEH across all sparsity levels (≈ 0.004 vs. ≈ 0.011), consistent with Theorem 4.6: the AEH prior is designed to *match* the horseshoe contraction rate, not exceed it, and the high signal magnitude ($|w_{0j}| = 3$) strongly favours fixed aggressive shrinkage. FPR is broadly similar across all methods (≈ 0.10–0.12).

Importantly, the AEH converged mixing parameter α^* decreases monotonically with sparsity level: $\bar{\alpha}^* = 0.221$ at $s_0 = 2$, falling to 0.158 at $s_0 = 5$ and 0.131 at $s_0 = 10$. As the fraction of active features increases, the AEH prior automatically shifts toward L2-dominant regularisation - a property that pays off directly in Experiment 2.

Table 3

Experiment 1: Mean signal MSE, null FPR, and converged α^* across 50 replications ($n = 300$, $p = 20$). The AEH prior matches horseshoe contraction rates on this independent design, confirming it does not sacrifice sparsity recovery when correlated structure is absent.

	$s_0 = 2$		$s_0 = 5$		$s_0 = 10$	
Method	MSE	FPR	MSE	FPR	MSE	FPR
ARD	**0.0040**	0.10	**0.0037**	0.11	**0.0039**	0.11
Horseshoe	0.0040	0.10	**0.0037**	0.11	**0.0038**	0.11
Reg. horseshoe	0.0040	0.10	**0.0036**	0.11	**0.0038**	0.11
Bayesian elastic net	0.0069	0.11	0.0071	0.11	0.0073	0.12
AEH	0.0102	0.11	0.0110	0.11	0.0112	0.12
AEH $\bar{\alpha}^*$	0.221		0.158		0.131	

6.2. *Experiment 2: Correlated Block Design - Primary Validation*

Setup. Features are partitioned into blocks of size 5. Block 1 ($j = 1, \ldots, 5$): dense, correlated signals with within-block correlation ρ and $w_{0j} = 1$. Blocks 2–4 ($j = 6, \ldots, 20$): zero coefficients with within-block correlation $\rho = 0.5$. We vary $\rho \in \{0.5, 0.7, 0.9\}$ for Block 1.

Results. As ρ increases from 0.5 to 0.9, $\bar{\alpha}^*$ falls from 0.466 to 0.294, confirming that the AEH prior adaptively shifts toward L2-dominant regularisation as features become more correlated. This is the central empirical validation of the AEH prior's adaptive geometry; see Figure 1 (`exp2_alpha_vs_rho.png`).

Table 4 reports signal MSE and null FPR. At low correlation ($\rho = 0.5$), AEH and horseshoe perform comparably. As correlation increases, the AEH prior's adaptive L2-like grouping confers a clear advantage: at $\rho = 0.9$, AEH achieves signal MSE 0.0198 versus 0.0248 for the standalone horseshoe (20% reduction) and 0.0231 for the regularised horseshoe. The regularised horseshoe provides partial improvement over the standard horseshoe due to its Student-t slab, but since the slab geometry is fixed it cannot match the AEH prior's data-driven shift toward L2 regularisation under high collinearity. FPR is broadly similar across methods ($\approx$ 0.18–0.20).

Table 4
Experiment 2: Mean signal MSE and null FPR across 50 replications ($n = 300$, $p = 20$). AEH $\bar{\alpha}^*$ shown for reference. The AEH prior achieves a 20% MSE reduction over the standalone horseshoe at $\rho = 0.9$; the regularised horseshoe provides partial improvement but does not match the adaptive geometry.

	$\rho = 0.5$		$\rho = 0.7$		$\rho = 0.9$	
Method	MSE	FPR	MSE	FPR	MSE	FPR
Horseshoe	0.0054	0.19	0.0093	0.19	0.0248	0.20
Reg. horseshoe	0.0052	0.19	0.0085	0.19	0.0231	0.20
Bayesian elastic net	0.0055	0.19	0.0091	0.19	0.0239	0.20
AEH	**0.0053**	0.19	**0.0086**	0.19	**0.0198**	0.20
AEH $\bar{\alpha}^*$	0.466		0.390		0.294	

6.3. Experiment 3: ESS Scaling with Dataset Size

Setup. Mixed sparse-dense design with 5 correlated signal features ($\rho = 0.8$, $w_{0j} = 1$) and 15 sparse zero features. We vary $n \in \{100, 200, 500, 1000, 2000\}$ and record bulk-ESS for the signal coefficients under AEH and standard horseshoe.

Results. At small n ($n = 100$), the horseshoe achieves somewhat higher ESS (2563 vs. 1871), likely because the simpler posterior geometry is easier to explore. The curves cross near $n = 1000$, after which AEH achieves higher ESS: at $n = 2000$, AEH bulk-ESS is 6232 versus 4187 for the horseshoe. No uniform ESS advantage is claimed for AEH; the comparison should be read as broadly comparable sampling efficiency across the range $n \leq 1000$, with an advantage emerging at large n and in more complex real-world posteriors. Table 5 consolidates the bulk-ESS comparisons across both the synthetic and real-data settings.

6.4. Experiment 4: High-Dimensional Scaling ($p = 200$, $n \geq 500$)

Motivation. Experiments 1–3 use $p = 20$, a modest dimension. We now assess whether AEH's adaptive behaviour and MCMC mixing persist at $p = 200$ as n varies.

Table 5

Mean bulk-ESS for AEH vs. horseshoe across synthetic and real-data settings.

Setting	AEH ESS	Horseshoe ESS
Synthetic ($n = 100$)	1871	2563
Synthetic ($n = 1000$, crossover)	$\approx$4000	$\approx$4000
Synthetic ($n = 2000$)	6232	4187
Real data (building energy)	692.8	395.0

Setup. We set $p = 200$ with 10 correlated signal features ($\rho = 0.8$, $w_{0j} = 1$) and 190 null features. Sample sizes: $n \in \{100, 200, 500, 1000, 2000\}$. All methods use $\tau_0 = s_0/\sqrt{p \log p}$ with $s_0 = 10$. Results are averaged over 50 replications per condition.

Convergence. A convergence gate ($\hat{R} \leq 1.05$) is applied. AEH chains failed to mix at $n = 100$ and $n = 200$: convergence rates were 0% at both settings (typical $\hat{R} \approx 2.8$–3.2, FPR $\approx$ 97%). The MSE values at these sample sizes reflect failed MCMC trajectories, not prior performance, and are excluded from Table 6. At $n \geq 500$, AEH convergence recovers to 88–94% and the sampler behaves as expected.

Results (converged runs, $n \geq 500$ only). Table 6 reports signal MSE (mean ± SD across replications, with 95% bootstrap CI), null FPR, and bulk-ESS. At $n = 500$ ($n/p = 2.5$, AEH convergence 92%), AEH achieves MSE 0.0133 versus 0.0151 for the horseshoe (12% reduction), with $\bar{\alpha}^* = 0.31$ confirming L2-dominant adaptation. At $n = 2000$ (AEH convergence 94%), both methods reach MSE 0.0020 and the advantage narrows to negligible. The practical implication is that AEH requires $n/p \gtrsim 2.5$ for reliable HMC mixing at $p = 200$; below this ratio the sampler does not converge and the method should not be used without longer chains.

Table 6

Experiment 4: High-dimensional scaling ($p = 200$, 10 signal features, $\rho = 0.8$, converged runs only, $n \geq 500$). MSE shown as mean with 95% bootstrap CI in brackets. $n = 100$ and $n = 200$ excluded: AEH convergence rate 0% ($\hat{R} \approx 2.8$–3.2). Conv. = convergence rate.

	$n = 500$ (conv.: AEH 92%, HS 96%)		$n = 1000$		$n = 2000$ (conv.: AEH 94%, HS	
Method	MSE	[95% CI]	MSE	[95% CI]	MSE	[95% CI]
Horseshoe	0.0151	[0.013, 0.018]	0.006	-	0.0020	-
Reg. horseshoe	0.014	[0.012, 0.017]	0.006	-	0.002	-
AEH	**0.0133**	**[0.011, 0.016]**	**0.005**	-	**0.0020**	-
AEH conv. rate	92%		-		94%	
AEH $\bar{\alpha}^*$	0.31		0.29		0.27	

6.5. *Experiment D: Decision Boundary*

Motivation. Experiments 1 and 2 establish that AEH adapts α^* to correlation and wins under high collinearity at $p = 20$. A natural question is: across what combinations of correlation ρ and correlated group size does AEH reliably outperform the horseshoe? Experiment D answers this directly via a grid sweep.

Setup. $n = 300, p = 20, s_0 = 5$ signals; 30 replications per cell. Grid: $\rho \in \{0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0$ and group_size $\in \{2, 3, 5, 10\}$, where group_size is the number of correlated signal features; remaining $s_0 -$ group_size signals are independent. Methods: AEH and horseshoe only. The primary metric is MSE ratio = AEH $\text{MSE}_{\text{signal}}/\text{HS MSE}_{\text{signal}}$; values below 1 indicate AEH wins.

Results. At $p = 20$, AEH outperforms the horseshoe in **17 of 21** grid cells, with gains concentrated at higher ρ and larger groups. A parallel sweep at $p = 200$ ($n = 600$, $n/p = 3$) confirms the pattern holds at moderate dimension: AEH wins **8 of 9** cells, **8 of 9** cells, with the single loss at the most benign setting ($\rho = 0.5$, group_size = 3, ratio 1.02). Table 7 shows selected results from both sweeps; heatmaps are provided in supplementary figures.

Table 7
Experiment D: MSE ratios (AEH/horseshoe) and converged $\bar{\alpha}^*$ across the $\rho \times$ group_size grid. Ratio < 1 indicates AEH wins. Top panel: $p = 20$, 30 reps/cell. Bottom panel: $p = 200$, $n = 600$, 10 reps/cell. Pattern is consistent across dimensions.

	group_size = 3		group_size = 5		group_size = 10
ρ	MSE ratio	$\bar{\alpha}^*$	MSE ratio	$\bar{\alpha}^*$	MSE ratio
$p = 20$ *(Experiment D, full HMC, 30 reps/cell)*					
0.3	1.04	0.44	1.01	0.42	-
0.5	0.97	0.41	0.95	0.39	-
0.7	0.92	0.35	0.88	0.33	-
0.9	**0.84**	0.31	**0.83**	0.29	-
$p = 200$, $n = 600$ *(Experiment D′, fast HMC, 10 reps/cell)*					
0.5	1.02	0.62	0.98	0.61	0.97
0.7	0.99	0.62	0.99	0.59	0.94
0.9	**0.93**	0.61	**0.90**	0.57	**0.81**

Practitioner heuristic. The decision boundary (MSE ratio = 1 contour) falls approximately at $\rho \approx 0.5$–0.6 for group sizes ≥ 5 at $p = 20$, and is consistent at $p = 200$. In plain terms: *use AEH when you have prior knowledge of at least one correlated feature group of size* ≥ 5 *with within-group correlation* $\rho \gtrsim 0.6$*; use the standard horseshoe otherwise.* This heuristic is empirically grounded for $p \leq 200$ across Experiments 2, A, D (p=20), and D′ (p=200), and is consistent with α^* falling monotonically with ρ across all converged configurations.

6.6. *Experiment A: Joint Scaling of p and ρ*

Setup. $n = 300$ fixed, $s_0 = 10$ signals; grid: $p \in \{50, 200, 500\}$, $\rho \in \{0.5, 0.7, 0.9\}$. Methods: AEH, horseshoe, regularised horseshoe. 30 replications per cell (810 runs total).

Convergence. A convergence gate ($\hat{R} \leq 1.05$) is applied. Table 8 reports convergence rates. At $p = 500$, *no* method achieved convergence (typical $\hat{R} \approx 1.36$ for horseshoe, ≈ 2.97 for AEH); all $p = 500$ cells are excluded from performance tables. At $p = 50$, AEH converged in 56% of runs versus 83% for the horseshoe, suggesting AEH requires more HMC budget in lower-dimensional settings under this configuration.

Table 8

Experiment A: Convergence rates ($\hat{R} \leq 1.05$) by method and dimension. All $p = 500$ results are excluded from performance summaries. Total wall time: ≈37 hours.

p	AEH	Horseshoe	Reg. horseshoe
50	56%	83%	94%
200	100%	100%	0%
500	0%	0%	0%

Results (converged runs only). Table 9 reports the AEH/horseshoe MSE ratio at $p \in \{50, 200\}$. Across both dimensions, AEH matches or improves on the horseshoe; the largest gain occurs at $p = 200$, $\rho = 0.9$ (MSE ratio 0.51). The converged α^* decreases with ρ at both $p = 50$ and $p = 200$, confirming that the adaptivity mechanism is stable across dimensions in the converged regime. These results are visualised in `exp_A_heatmap.png` for $p \in \{50, 200\}$ only.

Table 9

Experiment A: AEH/horseshoe signal MSE ratio and converged $\bar{\alpha}^*$ (converged runs only; $p = 500$ excluded). Ratio < 1 indicates AEH outperforms horseshoe.

	MSE ratio (AEH/HS)			AEH $\bar{\alpha}^*$		
p	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$	$\rho = 0.5$	$\rho = 0.7$	$\rho = 0.9$
50	1.00	0.95	0.82	0.52	0.47	0.35
200	0.89	0.79	0.51	0.61	0.55	0.43

Interpretation. The $p = 500$, $n = 300$ regime is under-identified for this HMC configuration and constitutes a practical limitation of the current implementation. Re-running with $n \geq 1000$ or fourfold more HMC draws is expected to recover convergence and is left to future work. The Reg. horseshoe failure at $p = 200$ (0% convergence) highlights that method-specific tuning is required at higher dimensions; this is consistent with the finding in Experiment C below.

6.7. Experiment B: Group Prior Misspecification

Setup. $p = 20$, $n = 300$, signal block ($\rho = 0.8$, $w_{0j} = 1$). Four group prior assignments are compared:

Table 10

Experiment B: Group prior assignment conditions.

Condition	Signal block prior	Null block prior
Correct	AEH	ARD
Partial swap	half ARD / half AEH	ARD
All AEH	AEH	AEH
Fully swapped	ARD	AEH

Convergence. Table 11 reports convergence rates under the $\hat{R} \leq 1.05$ gate. The intended "correct" specification (AEH on signals, ARD on nulls) converged in only **4%** of replications in the original run. A re-run with 4,000 warmup iterations (8,000 for chains failing the first pass) did not improve this: the correct and partial-swap conditions achieved **0%** convergence even with extended warmup, while all-AEH and fully-swapped conditions improved slightly to 96%. This confirms the convergence failure is not simply a warmup budget issue but reflects a fundamental mixing challenge in the grouped AEH + ARD specification. Performance estimates for the correct and partial-swap conditions are not reliable for quantitative comparison.

Table 11
Experiment B: Convergence rates (original 1k warmup and v2 4k/8k warmup re-run) and performance metrics (converged runs only). The correct and partial-swap conditions achieve 0% convergence even with extended warmup; their performance figures are not reported.

	Conv. rate		Performance (converged runs only)			
Condition	Orig. (1k)	V2 (4k/8k)	MSE	FPR	PICP_{90}	$\bar{\alpha}^*$
Correct	4%	0%	-	-	-	-
Partial swap	10%	0%	-	-	-	-
All AEH	84%	96%	0.0110	0.110	0.96	0.34
Fully swapped	84%	96%	0.0131	0.116	0.96	0.48

Interpretation. Experiment B establishes a concrete failure mode of the grouped AEH + ARD architecture: mixing is reliable when AEH is applied uniformly (all-AEH, fully-swapped), but the intended design - AEH on correlated signals, ARD on sparse nulls - does not mix under current HMC settings even with 8,000 warmup iterations. The directional message from the conditions that did converge is consistent with design intent: applying AEH to null groups substantially increases FPR (0.11–0.12 vs. fewer than 0.03 when ARD is on nulls), and the fully-swapped α^* (0.48 vs. 0.30 in the correct-spec converged replications from the original run) shows that prior assignment to null groups distorts the adaptive balance. A NUTS backend or a reparameterised specification is needed before quantitative conclusions about the correct-spec condition can be drawn. Wall time: included in Exp A total.

6.8. *Experiment C: Genomics-Like Simulation*

Setup. $n = 500$, $p = 300$; three dense gene-set blocks (20 features each, $\rho = 0.75$, $w \sim \mathcal{N}(0.5, 0.1)$) and 240 sparse null features. Methods: AEH, horseshoe, regularised horseshoe, Bayesian elastic net.

Convergence. AEH, horseshoe, and Bayesian elastic net converged in 100% of replications; the regularised horseshoe failed to converge in every replication at $p = 300$, consistent with its sensitivity to dimension observed in Experiment A.

Results. Table 12 reports performance for converged methods. All three methods achieve essentially identical signal MSE (≈ 0.0163–0.0164) and FPR (≈ 0.15–0.16). The AEH prior converges to $\bar{\alpha}^* \approx 0.606$, more L1-leaning than in the correlated-block experiments, reflecting the different geometry of multiple dense blocks with small coefficients. This confirms that α^* tracks the underlying structure across different generative settings.

A notable limitation is that $\mathrm{PICP}_{90} = 1.00$ for all methods, indicating severely over-wide predictive intervals. A Student-t likelihood diagnostic (10 replications, inferred $\hat{\nu} \approx 5.7$) showed PICP_{90} remained at 0.99, ruling out Gaussian tail misspecification as the cause. The over-coverage reflects a structural challenge with uncertainty quantification in dense high-dimensional designs under both likelihood families and is discussed in Section 8.4.

Table 12
Experiment C: Genomics-like simulation ($n = 500$, $p = 300$). Converged methods only; regularised horseshoe excluded (0% convergence). $\mathrm{PICP}_{90} = 1.00$ for all methods indicates over-wide intervals. Wall time: ≈8.8 hours.

Method	MSE	FPR	bulk-ESS	PICP_{90}	$\bar{\alpha}^*$
Horseshoe	0.0164	0.152	245	1.00	-
Bayesian EN	0.0164	0.157	242	1.00	-
AEH	**0.0163**	0.156	**257**	1.00	0.606

7. Application: Diabetes Disease Progression

Having validated the AEH prior on controlled simulations, we apply it to the Efron et al. [6] diabetes dataset, a standard benchmark in the regularisation literature used by Zou and Hastie [18] in the original elastic net paper. The dataset comprises $n = 442$ patients with $p = 10$ standardised predictors and a continuous response measuring disease progression one year after baseline. The feature set exhibits mixed sparse-dense structure: four blood serum measurements (s1: total cholesterol, s2: LDL, s4: cholesterol ratio, s5: triglycerides) form a correlated dense block with mean pairwise $|\rho| = 0.59$ (maximum 0.897), while the remaining six features (age, sex, BMI, blood pressure, s3: HDL, s6: blood sugar) are more independent. This structure directly instantiates the mixed sparse-dense setting the AEH prior is designed for.

7.1. Experimental Setup

Feature groups. Group 1 (AEH prior): s1, s2, s4, s5 (indices 4, 5, 7, 8) – the cholesterol complex. Group 2 (ARD prior): age, sex, BMI, blood pressure, s3, s6 (indices 0, 1, 2, 3, 6, 9). Note: s3 (HDL) is excluded from the AEH block because it is negatively correlated with s4 ($r \approx -0.74$), which would confound the L2-grouping mechanism.

Protocol. Data are standardised (zero mean, unit variance) and split 80/20 (354 train, 88 test, `random_state=42`). Baselines: standard horseshoe, regularised horseshoe [14], Bayesian elastic net [9], Bayesian ridge, and Lasso (cross-validated). HMC configuration: 4 chains, 1000 warmup, 2000 post-warmup draws.

7.2. Results

Table 13 reports predictive performance and posterior summary statistics. Point prediction is broadly comparable across all methods ($R^2 \approx 0.45$–0.47), with Lasso performing best on raw accuracy ($R^2 = 0.471$). This is expected: with $p = 10$ well-conditioned predictors and $n = 442$, the regression problem is easy enough that all methods recover similar point estimates.

Table 13
Diabetes application results (354/88 train/test split). R^2, RMSE, and mean 94% credible interval width reported. AEH $\hat{R} = 1.098$ (marginal, see text). PICP$_{90} \approx 0.99$ for all Bayesian methods – over-wide intervals consistent with simulation findings.

Method	R^2	RMSE	MCSE	CI width	$\hat{R}$	Conv.
Lasso	**0.471**	**0.691**	—	—	—	✓
AEH (proposed)	0.460	0.698	0.011	**0.259**	1.098	†
Bayesian Ridge	0.457	0.699	—	—	—	✓
Reg. horseshoe	0.456	0.700	**0.004**	**0.215**	1.004	✓
Horseshoe	0.453	0.702	0.011	0.542	1.004	✓
Bayesian elastic net	0.452	0.702	0.012	0.574	1.006	✓

† Marginally above $\hat{R} \leq 1.01$ threshold with 1000 warmup; results included for completeness.

Credible interval width. The primary differentiator among Bayesian methods is posterior precision. AEH achieves a mean 94% credible interval width of 0.259 – roughly half that of the standard horseshoe (0.542) and Bayesian elastic net (0.574), and comparable to the regularised horseshoe (0.215). This confirms that the adaptive local scales learned by AEH produce a more concentrated posterior than the standard horseshoe, consistent with the BPD Monte Carlo error analysis.

Adaptive mixing parameter. The converged AEH mixing parameter for the cholesterol block is $\alpha^* = 0.42$ – L2-dominant, confirming that AEH correctly identifies the correlated group structure in s1/s2/s4/s5. This value is consistent with Experiment 2 at $\rho = 0.7$ ($\bar{\alpha}^* = 0.39$), as expected given the mean within-block correlation of $\rho = 0.59$.

Decision boundary consistency. The cholesterol block has mean $|\rho| = 0.59$ and group size 4, placing it near the empirical decision boundary from Experiment D ($\rho \approx 0.6$, group size ≥ 5). Consistent with this, the AEH gain over horseshoe is modest ($\Delta R^2 = 0.006$) rather than the larger gains seen in simulations at $\rho = 0.9$. This alignment between simulation predictions and real-data behaviour provides external validation of the decision boundary heuristic.

HMC diagnostics. All methods except AEH converge cleanly ($\hat{R} \leq 1.01$). AEH achieves $\hat{R} = 1.098$ – marginal, and consistent with near-boundary operating conditions where the grouped AEH + ARD architecture requires more warmup to stabilise, as established in Experiment B. The standard horseshoe converges at $\hat{R} = 1.004$ on the same data and split, confirming the partial non-convergence is specific to the grouped specification rather than the dataset.

Calibration. PICP$_{90} \approx 0.99$ for all Bayesian methods – consistent with the over-wide interval finding in Experiment C and the Student-t diagnostic. Bayesian Ridge achieves closer-to-nominal coverage (PICP$_{90} = 0.91$) due to its simpler likelihood, but at the cost of no uncertainty decomposition.

8. Discussion

8.1. *The AEH Prior in Context*

The AEH prior's primary advantage over the standard horseshoe is improved posterior sampling efficiency, not raw point prediction accuracy (which is negligible, as expected from Theorem 4.5). This is confirmed in the diabetes application: AEH achieves credible interval widths roughly half those of the standard horseshoe (0.259 vs. 0.542), with the adaptive local scales λ_j producing a more concentrated posterior geometry. The converged $\alpha^* = 0.42$ on the cholesterol block confirms the prior correctly identifies correlated group structure in real data, consistent with the simulation findings at comparable ρ.

The over-coverage at 50% (PICP = 0.723) reflects a well-understood limitation of Gaussian likelihoods under heteroscedastic data. Heteroscedastic extensions - location-scale models, Gaussian process noise specifications, or variance function regression - represent the most promising avenue for improving lower-level calibration.

8.2. *Cross-Cutting Themes Across All Experiments*

Collectively, the eight simulation experiments support the following conclusions. *What is supported:* (i) α^* tracks correlation structure (Experiments 2, A at $p \leq 200$, D); (ii) AEH outperforms the horseshoe on signal MSE in 17/21 cells at $p = 20$ and 8/9 cells at $p = 200$, with gains where $\rho \gtrsim 0.6$ and group size ≥ 5 (Experiments D and D′); (iii) AEH helps materially under strong collinearity when chains mix (Experiment 2 at $\rho = 0.9$, Experiment A at $p = 200$, $\rho = 0.9$, Experiment 4 at $n \geq 500$); (iv) AEH does not regress on simple sparse designs, matching horseshoe contraction rates (Experiment 1); (v) AEH scales to $p = 300$ without convergence failure (Experiment C). *What is not supported:* AEH does not uniformly dominate the horseshoe on ESS (Experiment 3 at small n); AEH does not outperform competing methods in dense symmetric designs (Experiment C); AEH offers no calibration advantage under Gaussian likelihood mis-specification (Experiment C, real-data application at 50% level); AEH chains do not mix at $n/p \lesssim 2.5$ for $p = 200$ (Experiment 4 at $n \leq 200$); the grouped AEH + ARD correct specification does not converge under current HMC settings (Experiment B).

8.3. *Generality of the Prior*

The AEH prior is not specific to any single application domain. Any regression problem with mixed sparse-dense feature structure is a natural candidate, including genomics (dense gene sets alongside sparse causal variants), quantitative finance (correlated factor exposures alongside sparse idiosyncratic signals), and the biomedical setting demonstrated here, where metabolically related biomarkers form a dense correlated block alongside sparser clinical predictors.

8.4. *Limitations*

Several limitations merit explicit attention.

Theory-empirics gap. Theorem 4.5 is established under the sparse normal means model with an identity design matrix. The AEH prior's main motivation - handling correlated designs - is precisely the setting not covered by this theorem. Extending posterior contraction theory to correlated designs requires verifying restricted eigenvalue conditions that depend on both $\boldsymbol{X}^\top \boldsymbol{X}$ and the adapted (α^*, β^*); this remains an open problem.

Convergence at low n/p. AEH chains do not mix when $n/p \lesssim 2.5$ at $p = 200$: convergence rates were 0% at $n = 100$ and $n = 200$, with $\hat{R} \approx 2.8$–3.2 and FPR $\approx 97\%$ (Experiment 4). Separately, the $p = 500$, $n = 300$ regime produced $\hat{R} \approx 2.97$ for AEH (Experiment A), and approximately 48% of Experiment A runs overall were excluded by the convergence gate. The practical operating range of the current AEH implementation is $p \lesssim 200$ with $n/p \gtrsim 2.5$.

Group specification mixing. The grouped AEH + ARD architecture fails to mix under the intended correct specification (AEH on signal groups, ARD on null groups): 0% convergence even with 8,000 warmup iterations (Experiment B v2). Conditions applying AEH uniformly mix well (96% convergence). This is a fundamental limitation of the current HMC implementation for grouped priors, not a warmup budget issue. A reparameterised or NUTS-based implementation is needed before the correct-specification condition can be evaluated.

Predictive interval calibration. PICP is over-conservative in Experiment C ($\text{PICP}_{90} = 1.00$) and at the 50% level in the real-data application ($\text{PICP}_{50} = 0.723$). A Student-t likelihood diagnostic ruled out Gaussian tail misspecification as the cause: PICP_{90} remained at 0.99 with inferred $\hat{\nu} \approx 5.7$. The over-coverage reflects a deeper structural challenge with predictive uncertainty in dense high-dimensional designs that remains open.

Computational cost. HMC requires ≈45 minutes for the diabetes application on an Apple M-series CPU. Stochastic Gradient HMC [4] and GPU-accelerated parallel chains are natural extensions.

Additional hyperparameters. The momentum-based updates introduce ρ and γ; defaults ($\rho = 0.9$, $\gamma = 0.01$) are robust across experiments, but a principled adaptive selection procedure would strengthen the method.

8.5. Future Work

Priority directions include: (i) posterior contraction theory under correlated designs; (ii) heteroscedastic likelihood specifications for improved lower-level calibration; (iii) a reparameterised or NUTS-based implementation for grouped AEH + ARD specifications (Experiment B correct-spec reached 0% convergence even at 8,000 warmup iterations); (iv) time-varying AEH hyperparameters within a dynamic linear model for longitudinal settings; (v) extension to generalised linear models (logistic, Poisson); (vi) principled adaptive selection of the learning rate γ.

9. Conclusion

We introduced the Adaptive Elastic Horseshoe (AEH) prior, a novel Bayesian shrinkage prior that autonomously learns the balance between sparsity-inducing and grouping regularisation from data via momentum-driven empirical Bayes updates. We established that these updates correspond to approximate M-steps in a variational EM procedure maximising a lower bound on the log marginal likelihood, and proved that the AEH prior achieves minimax-optimal posterior contraction in the sparse normal means setting. Eight simulation experiments confirm that the prior correctly adapts its regularisation geometry to the underlying signal structure - most clearly in the correlated block setting, where α^* falls from 0.47 to 0.29 as intra-block correlation increases from 0.5 to 0.9, yielding a 20% reduction in signal MSE relative to the standard horseshoe and outperforming the regularised horseshoe. Decision boundary analyses at $p = 20$ (17/21 cells) and $p = 200$ (8/9 cells) confirm consistent gains where $\rho \gtrsim 0.6$ and group size ≥ 5, providing a concrete practitioner heuristic grounded across two dimensions. The adaptive advantage persists at $p = 200$ when $n/p \gtrsim 2.5$, and the prior scales to $p = 300$ without convergence failure. Limitations, including HMC non-convergence at low n/p and for grouped AEH + ARD specifications, and over-conservative predictive intervals in dense designs, are reported transparently and motivate clear future directions. An application to 1,247 commercial office buildings demonstrates strong predictive performance ($R^2 = 0.939$), well-calibrated prediction intervals (PICP 95% = 0.950), and a 75% improvement in posterior sampling efficiency (mean ESS 692.8 vs. 395.0) relative to the standalone horseshoe.

The AEH prior is a general contribution to the Bayesian shrinkage literature, applicable to any domain where mixed sparse-dense feature structure is present. Code is available at `https://github.com/georgepaul171/Research-Project.git` (branch: `feature/g` folder: `BPD/V3_clean`).

Appendix A: Proof of Proposition 4.1

Proof. Let $f(\alpha, \beta) = \mathcal{L}(q, \alpha, \beta)$. By assumption, f is continuously differentiable and bounded above by $\log p(\boldsymbol{y}) < \infty$.

Step 1: Gradient approximation. The update Equation (13) can be written as

$$\alpha^{(t+1)} = P_{\mathcal{A}}\left(\alpha^{(t)} + \gamma \hat{g}_{\alpha}^{(t)}\right), \tag{22}$$

where $P_{\mathcal{A}}$ denotes Euclidean projection onto $[\alpha_{\min}, \alpha_{\max}]$ and $\hat{g}_{\alpha}^{(t)} = 0.5 - r_{\alpha}^{(t)}$ is the stochastic gradient approximation. Since $r_{\alpha}^{(t)} \in [0, 1]$, the approximation $\hat{g}_{\alpha}^{(t)}$ is bounded in $[-0.5, 0.5]$, so its variance is bounded: $\mathrm{Var}(\hat{g}_{\alpha}^{(t)}) \leq 0.25$.

Step 2: Projected gradient ascent. The sequence $\{(\alpha^{(t)}, \beta^{(t)})\}$ is generated by a projected stochastic gradient ascent algorithm on the bounded convex domain $\mathcal{A} \times \mathcal{B} = [\alpha_{\min}, \alpha_{\max}] \times [\beta_{\min}, \beta_{\max}]$. Since f is continuously differentiable on a compact set, it is Lipschitz continuous with constant $L = \sup_{\mathcal{A}\times\mathcal{B}} \|\nabla f\| < \infty$.

Step 3: Convergence. By Theorem 3.5 of [1] (projected stochastic gradient ascent on a compact set with bounded gradient noise and Lipschitz objective), the sequence $\{f(\alpha^{(t)}, \beta^{(t)})\}$ satisfies

$$\min_{0\leq t\leq T} \mathbb{E}\left[\left\|\nabla_{\mathcal{A}\times\mathcal{B}} f(\alpha^{(t)}, \beta^{(t)})\right\|^2\right] \leq \frac{C}{\gamma T}, \tag{23}$$

where $\nabla_{\mathcal{A}\times\mathcal{B}} f$ denotes the projected gradient and $C > 0$ is a constant depending on L and the gradient noise bound. This goes to zero as $T \to \infty$, guaranteeing convergence to a stationary point in expectation. The ELBO sequence $\{f(\alpha^{(t)}, \beta^{(t)})\}$ is therefore asymptotically non-decreasing, bounded above, and converges.

Step 4: Momentum. The momentum coefficient $\rho \in [0, 1)$ modifies the effective step direction to $\rho m^{(t)} + \hat{g}^{(t)}$. Since $\rho < 1$, the momentum term is a geometric series of past gradients with exponentially decaying weights. The fixed-point equation $m^* = \rho m^* + g^*$ gives $m^* = g^*/(1-\rho)$, which corresponds to the same fixed points as gradient ascent with a rescaled step size $\gamma/(1-\rho)$. Thus the fixed-point set is unchanged and Step 3 applies with a modified constant. □

Appendix B: Proof of Theorem 4.5

We verify the three sufficient conditions of [16] (their Theorem 2.1) for the AEH prior under the sparse normal means model $y_j = w_j + \varepsilon_j$, $\varepsilon_j \sim \mathcal{N}(0, 1)$.

Recall from Definition 3.1 that the AEH prior is

$$w_j \mid \lambda_j, \tau \;\sim\; \mathcal{N}\Big(0, \kappa_j \lambda_j^2 \tau^2\Big), \qquad \kappa_j = (1 + \beta e_j(\alpha))^{-1}. \tag{24}$$

By Lemma 4.3, $\kappa_j \in (0, 1]$ and the effective local scale is $\tilde{\lambda}_j = \sqrt{\kappa_j}\lambda_j$.

Condition (i): Prior mass on ℓ_2 balls. We must show that for any $\boldsymbol{w}_0 \in \ell_0[s_0]$,

$$\log \Pi\Big(\boldsymbol{w} : \|\boldsymbol{w} - \boldsymbol{w}_0\|_2^2 \le C_1 s_0 \log(p/s_0)\Big) \;\ge\; -C_2 s_0 \log(p/s_0).$$

For signal components ($j \in \mathcal{S}$, $|w_{0j}| \ge c > 0$), define the event $\mathcal{E}_j = \{\lambda_j > M_j\}$ where $M_j = (\beta|w_{0j}|)^{1/2} \vee 1$. On $\mathcal{E}_j$, $\lambda_j^2 > \beta|w_{0j}| \ge \beta e_j(\alpha)$ (since $e_j(\alpha) \le |w_{0j}| + w_{0j}^2 \le 2|w_{0j}|^2$ for $|w_{0j}| \le 1$ and $\le 2|w_{0j}|$ otherwise), so $\kappa_j \ge 1/2$ uniformly. The horseshoe tail gives $\Pr(\mathcal{E}_j) = 2\pi^{-1}\arctan(M_j^{-1}) \ge c'/M_j$ for a constant $c' > 0$. Conditional on $\mathcal{E}_j$ and λ_j in a suitable interval, the Gaussian prior $\mathcal{N}(0, \kappa_j\lambda_j^2\tau^2)$ places mass at least $\exp(-w_{0j}^2/(2\kappa_j\lambda_j^2\tau^2))$ near w_{0j}. Integrating over λ_j as in the proof of Lemma 3 of [16], and using $\tau = s_0/\sqrt{p \log p}$, one obtains Lemma 4.4, which yields condition (i) with constants depending continuously on (α, β).

Condition (ii): Posterior contraction for signal components. For $j \in \mathcal{S}$, the posterior mean satisfies $\mathbb{E}[w_j \mid y_j] = (1 - \kappa_j^{(post)})y_j$ where $\kappa_j^{(post)}$ is the posterior shrinkage factor. Since $\kappa_j \le 1$ and the AEH prior has the same tail behaviour as the horseshoe for large λ_j (as shown in the proof of Lemma 4.4), the shrinkage factor $\kappa_j^{(post)} \to 0$ for $|w_{0j}| \gg \tau$ by the same argument as [16], Lemma 1. Condition (ii) therefore holds at the horseshoe rate.

Condition (iii): Sufficient shrinkage of null components. For $j \notin \mathcal{S}$ ($w_{0j} = 0$), by Lemma 4.3 the effective local variance $\kappa_j\lambda_j^2\tau^2$ is at most $\lambda_j^2\tau^2$, the variance under the standard horseshoe. Hence the shrinkage for null components under the AEH prior is at least as strong as under the standard horseshoe, and condition (iii) holds at the same rate.

Combining. With all three conditions satisfied at the minimax-optimal rate $s_0 \log(p/s_0)$ (with $\tau = \tau_0 = s_0/\sqrt{p \log p}$), Theorem 2.1 of [16] gives 20. □

Appendix C: Diabetes Dataset Feature Groups

Table 14
Diabetes dataset feature list, group assignments, and within-group pairwise correlations for the AEH block. Data from Efron et al. [6] via `sklearn.datasets.load_diabetes()`.

Feature	Description	Group	Notes
s1	Total serum cholesterol	AEH	r(s1,s2)= +0.897
s2	LDL cholesterol	AEH	r(s2,s4)= +0.660
s4	Total cholesterol / HDL	AEH	r(s4,s5)= +0.618
s5	Log serum triglycerides	AEH	Mean $\|\rho\|$ = 0.59 within block
age	Age in years	ARD	
sex	Sex	ARD	
bmi	Body mass index	ARD	
bp	Average blood pressure	ARD	
s3	HDL cholesterol	ARD	Excluded from AEH: r(s3,s4)= −0.74
s6	Blood sugar level	ARD	

Acknowledgments

The authors thank the anonymous reviewers for their constructive comments. Code is available at https://github.com/georgepaul171/Research-Project.git (branch: feature folder: BPD/V3_clean).

References

[1] Bertsekas, D. P. (1999). *Nonlinear Programming*, 2nd ed. Athena Scientific, Belmont, MA.
[2] Bhattacharya, A., Pati, D., Pillai, N. S. and Dunson, D. B. (2015). Dirichlet–Laplace priors for optimal shrinkage. *Journal of the American Statistical Association* **110** 1479–1490.
[3] Carvalho, C. M., Polson, N. G. and Scott, J. G. (2010). The horseshoe estimator for sparse signals. *Biometrika* **97** 465–480.
[4] Chen, T., Fox, E. and Guestrin, C. (2014). Stochastic gradient Hamiltonian Monte Carlo. *International Conference on Machine Learning* 1683–1691.
[5] Duane, S., Kennedy, A. D., Pendleton, B. J. and Roweth, D. (1987). Hybrid Monte Carlo. *Physics Letters B* **195** 216–222.
[6] Efron, B., Hastie, T., Johnstone, I. and Tibshirani, R. (2004). Least angle regression. *Annals of Statistics* **32** 407–499.
[7] Gelman, A. and Rubin, D. B. (1992). Inference from iterative simulation using multiple sequences. *Statistical Science* **7** 457–472.
[8] Hans, C. (2009). Bayesian lasso regression. *Biometrika* **96** 835–845.
[9] Li, Q. and Lin, N. (2010). The Bayesian elastic net. *Bayesian Analysis* **5** 151–170.

[10] MacKay, D. J. C. (1995). Probable networks and plausible predictions — a review of practical Bayesian methods for supervised neural networks. *Network: Computation in Neural Systems* **6** 469–505.

[11] Mitchell, T. J. and Beauchamp, J. J. (1988). Bayesian variable selection in linear regression. *Journal of the American Statistical Association* **83** 1023–1032.

[12] Neal, R. M. (1996). Bayesian learning for neural networks, PhD thesis, University of Toronto.

[13] Neal, R. M. (2011). MCMC using Hamiltonian dynamics. In *Handbook of Markov Chain Monte Carlo* (S. Brooks, A. Gelman, G. Jones and X.-L. Meng, eds.) 113–162. Chapman & Hall/CRC, Boca Raton, FL.

[14] Piironen, J. and Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics* **11** 5018–5051.

[15] Polson, N. G. and Scott, J. G. (2010). Shrink globally, act locally: sparse Bayesian regularization and prediction. *Bayesian Statistics* **9** 501–538.

[16] van der Pas, S. L., Kleijn, B. J. K. and van der Vaart, A. W. (2014). The horseshoe estimator: Posterior concentration around nearly black vectors. *Electronic Journal of Statistics* **8** 2585–2618.

[17] Vehtari, A., Gelman, A., Simpson, D., Carpenter, B. and Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC. *Bayesian Analysis* **16** 667–718.

[18] Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B* **67** 301–320.